

Example:

$$f(x, y) = x + y + \frac{1}{xy}, \quad x, y > 0$$

Critical points:

$$f_x = 1 - \frac{1}{x^2y}, \quad f_y = 1 - \frac{1}{xy^2}$$

Set them equal to zero:

$$1 - \frac{1}{x^2y} = 0 \quad \Rightarrow \quad x^2y = 1$$

$$1 - \frac{1}{xy^2} = 0 \quad \Rightarrow \quad xy^2 = 1$$

Solve the system:

$$x^2y = 1, \quad xy^2 = 1$$

Divide the equations:

$$\frac{x^2y}{xy^2} = \frac{1}{1} \Rightarrow \frac{x}{y} = 1 \Rightarrow x = y$$

Substitute:

$$x^3 = 1 \Rightarrow x = 1, \quad y = 1$$

Critical point: $(1, 1)$

Second derivatives:

$$f_{xx} = \frac{2}{x^3y}, \quad f_{yy} = \frac{2}{xy^3}, \quad f_{xy} = \frac{1}{x^2y^2}$$

At $(1, 1)$:

$$A = f_{xx} = 2, \quad B = f_{xy} = 1, \quad C = f_{yy} = 2$$

$$AC - B^2 = (2)(2) - (1)^2 = 3 > 0, \quad A > 0$$

$\Rightarrow$ Local minimum at $(1, 1)$

Behavior:

$$f \rightarrow \infty \text{ as } x \rightarrow \infty \text{ or } y \rightarrow \infty$$

$$f \rightarrow \infty \text{ as } x, y \rightarrow 0$$

Lecture: Differentials, Chain Rule

More tools to study functions.

Differentials

Implicit differentiation

$$y = f(x)$$

$$dy = f'(x) dx$$

Example:

$$y = \sin^{-1}(x)$$

$$x = \sin y$$

Differentiate both sides:

$$\frac{dx}{dy} = \cos(y)$$

$$\frac{dy}{dx} = \frac{1}{\cos(y)} = \frac{1}{\sqrt{1-x^2}}$$

Total Differential

For a function of three variables:

$$f(x, y, z)$$

$$df = f_x dx + f_y dy + f_z dz$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

Important:

$$df \neq \Delta f$$

Uses:

1. Encodes how changes in x, y, z affect f
2. Acts as a placeholder for small variations $\Delta x, \Delta y, \Delta z$ to approximate:

$$\Delta f \approx df$$

Chain Rule via Differentials

$$\Delta f \approx f_x \Delta x + f_y \Delta y + f_z \Delta z$$

Divide by Δt to get rate of change:

$$\frac{\Delta f}{\Delta t} \approx f_x \frac{\Delta x}{\Delta t} + f_y \frac{\Delta y}{\Delta t} + f_z \frac{\Delta z}{\Delta t}$$

As $\Delta t \rightarrow 0$:

$$\frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}$$

When:

$$x = x(t), \quad y = y(t), \quad z = z(t)$$

Why is this valid?

First attempt:

$$df = f_x dx + f_y dy + f_z dz$$

$$dx = x'(t) dt, \quad dy = y'(t) dt, \quad dz = z'(t) dt$$

Substitute:

$$df = f_x x'(t) dt + f_y y'(t) dt + f_z z'(t) dt$$

Factor out dt :

$$df = (f_x x'(t) + f_y y'(t) + f_z z'(t)) dt$$

Divide by dt :

$$\begin{aligned} \frac{df}{dt} &= f_x x'(t) + f_y y'(t) + f_z z'(t) \\ &= f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt} \end{aligned}$$

Limit Interpretation

$$\frac{\Delta f}{\Delta t} \rightarrow \frac{df}{dt}, \quad \frac{\Delta x}{\Delta t} \rightarrow \frac{dx}{dt} \quad \text{as } \Delta t \rightarrow 0$$

Example:

$$w = x^2 y + z, \quad x = t, \quad y = e^t, \quad z = \sin t$$

Chain Rule

$$\begin{aligned}
\frac{dw}{dt} &= 2xy \frac{dx}{dt} + x^2 \frac{dy}{dt} + \frac{dz}{dt} \\
&= 2te^t(1) + t^2e^t + \cos t \\
&= 2te^t + t^2e^t + \cos t
\end{aligned}$$

Substitute:

$$w = x^2y + z$$

$$w(t) = t^2e^t + \sin t$$

$$\frac{dw}{dt} = 2te^t + t^2e^t + \cos t$$

Application: Product Rule

$$f = uv, \quad u = u(t), \quad v = v(t)$$

$$\frac{d(uv)}{dt} = v \frac{du}{dt} + u \frac{dv}{dt}$$

Quotient Rule

$$g = \frac{u}{v}, \quad u = u(t), \quad v = v(t)$$

$$\begin{aligned}
\frac{d}{dt} \left(\frac{u}{v} \right) &= \frac{1}{v} \frac{du}{dt} - \frac{u}{v^2} \frac{dv}{dt} \\
&= \frac{v \frac{du}{dt} - u \frac{dv}{dt}}{v^2}
\end{aligned}$$

Chain Rule with more variables

$$w = f(x, y), \quad \text{where } x = x(u, v), \quad y = y(u, v)$$

$$f(x(u, v), y(u, v))$$

Question: Find $\frac{\partial w}{\partial u}, \frac{\partial w}{\partial v}$

$$\frac{\partial w}{\partial x}, \quad \frac{\partial w}{\partial y}, \quad \frac{\partial x}{\partial u}, \quad \frac{\partial x}{\partial v}, \quad \frac{\partial y}{\partial u}, \quad \frac{\partial y}{\partial v}$$

Total differential:

$$dw = f_x dx + f_y dy$$

$$\begin{aligned}
&= f_x \left(\frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv \right) + f_y \left(\frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv \right) \\
&= (f_x x_u + f_y y_u) du + (f_x x_v + f_y y_v) dv
\end{aligned}$$

$$\frac{\partial w}{\partial u} = f_x \frac{\partial x}{\partial u} + f_y \frac{\partial y}{\partial u}$$

$$\frac{\partial w}{\partial v} = f_x \frac{\partial x}{\partial v} + f_y \frac{\partial y}{\partial v}$$

Example: Polar Coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$f = f(x, y)$$

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r}$$

Gradient Vector

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

Gradient Directional (Chain Rule)

$$\frac{dw}{dt} = w_x \frac{dx}{dt} + w_y \frac{dy}{dt} + w_z \frac{dz}{dt}$$

$$\frac{d\mathbf{r}}{dt}$$

$$\nabla w = \langle w_x, w_y, w_z \rangle$$

Gradient and Level Surfaces

Gradient of w at some point

$$\frac{dx}{dt}, \quad \frac{dy}{dt}, \quad \frac{dz}{dt}$$

Theorem

∇w is normal to the level surface where $w = \text{constant}$.

Example 1

$$w = a_1x + a_2y + a_3z$$

$$\nabla w = \langle a_1, a_2, a_3 \rangle$$

Level surface:

$$a_1x + a_2y + a_3z = c$$

This represents a plane with normal vector $\langle a_1, a_2, a_3 \rangle$.

Example 2

$$w = x^2 + y^2$$

Level curve:

$$x^2 + y^2 = c$$

$$\nabla w = \langle 2x, 2y \rangle$$

∇w points towards higher values of w .

Proof

Take a curve $\vec{r}(t)$ that stays on the level surface $w = c$.

Proof (continued)

Let $w(x, y, z) = c$.

Let $\vec{v} = \frac{d\vec{r}}{dt}$ be tangent to the level surface $w = c$.

By the chain rule:

$$\begin{aligned}\frac{dw}{dt} &= \nabla w \cdot \frac{d\vec{r}}{dt} \\ &= \nabla w \cdot \vec{v}\end{aligned}$$

Since $w(t) = c$ (constant), we have:

$$\frac{dw}{dt} = 0$$

So:

$$\nabla w \cdot \vec{v} = 0$$

This means $\nabla w \perp \vec{v}$.

Since $\vec{v}$ is any vector tangent to the level surface, it follows that:

$$\nabla w \perp \text{tangent plane to the level surface}$$

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Example

Find the tangent plane to the surface:

$$x^2 + y^2 - z^2 = 4$$

at the point $(2, 1, 1)$.

Solution:

Let

$$w(x, y, z) = x^2 + y^2 - z^2$$

$$\nabla w = \langle 2x, 2y, -2z \rangle$$

At $(2, 1, 1)$:

$$\nabla w = \langle 4, 2, -2 \rangle$$

Equation of tangent plane:

$$4(x - 2) + 2(y - 1) - 2(z - 1) = 0$$

Simplified:

$$4x + 2y - 2z = 8$$

Continuation

Level set:

$$w = 4 \quad \text{where } w = x^2 + y^2 - z^2$$

Gradient:

$$\nabla w = \langle 2x, 2y, -2z \rangle$$

At $(2, 1, 1)$:

$$\nabla w = \langle 4, 2, -2 \rangle$$

So the equation is:

$$4x + 2y - 2z = 8$$

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Another way:

$$dw = 2x \, dx + 2y \, dy - 2z \, dz$$

At $(2, 1, 1)$:

$$dw = 4 \, dx + 2 \, dy - 2 \, dz$$

$$\Delta w \approx 4\Delta x + 2\Delta y - 2\Delta z$$

On the level surface, $\Delta w = 0$, so:

$$4\Delta x + 2\Delta y - 2\Delta z = 0$$

Equation of tangent plane:

$$4(x - 2) + 2(y - 1) - 2(z - 1) = 0$$

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Directional Derivatives

Let:

$$w = w(x, y)$$

We know:

$$\frac{\partial w}{\partial x}, \quad \frac{\partial w}{\partial y}$$

Derivatives in the direction $\mathbf{i}$ or $\mathbf{j}$.

What if we move in direction of a unit vector $\mathbf{u}$?

Consider the surface:

$$z = w(x, y)$$

Take a straight-line trajectory:

$$\vec{r}(s)$$

Then:

$$\frac{d\vec{r}}{ds} = \mathbf{u}$$

where s is arc length along the line.